Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no...2.5.9.k.0.1.2.............

Name of the Course: B.Tech.
Semester: II
Name of the Paper: Engineering Mathematics-II
Paper Code: TMA-201

Time: 1.5 Hours **Maximum Marks: 50**

Note:
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. (10 Marks) (CO1)

a. Solve: $\frac{dz}{dx}+\frac{z}{x}\log z=\frac{z}{x^2}(\log z)^2$.

OR

b. Solve: $\left(1+ae^{x/y}\right)dx+ae^{x/y}\left(1-\frac{x}{y}\right)dy=0$.

Q2. (10 Marks) (CO1)

a. Solve: $\frac{d^2y}{dx^2}-(a+b)\frac{dy}{dx}+aby=e^{ax}+e^{bx}$.

OR

b. Solve: $\left(x^3D^3+x^2D^2-2\right)y=x-\frac{1}{x^3}$.

Q3. (10 Marks)

a. Solve: $\frac{d^2y}{dx^2}-y=\frac{2}{1+e^x}$ by using variation of parameters method. (CO1)

OR

b. Solve the following differential equation in series: $\left(1-x^2\right)\frac{d^2y}{dx^2}-x\frac{dy}{dx}+4y=0$. (CO2)

Q4. (10 Marks) (CO2)

a. Prove that: $\int_{-1}^{1}P_m(x)P_n(x)\,dx=0$ if $m\neq n$.

OR

b. Prove that: $(2n+1)xP_n=(n+1)P_{n+1}+nP_{n-1}$.

Q5. (10 Marks) (CO2)

a. Prove that: $xJ_n'=nJ_n-xJ_{n+1}$.

OR

b. Prove that: $J_{3/2}=\sqrt{\frac{2}{\pi x}}\left[\frac{\sin x}{x}-\cos x\right]$.